5 (2026) 100013

Contents lists available at ScienceDirect

Brain and Environment

journal homepage: www.sciencedirect.com/journal/brain-and-environment

Associations between surrogates of prenatal lead DNA methylation and adolescent neural network architecture

Ryan Tung, M.S. [a,c], Carly O'Neill, B.S. [b], Trey Smith, B.S. [e,f], Felicia A. Hardi, Ph.D. [d], Leigh G. Goetschius, Ph.D. [i], Edward D. Huntley, Ph.D. [a,e], Helen C.S. Meier, Ph.D., M.P.H. [e,f], Luke W. Hyde, Ph.D. [a,e,h], Dan Notterman, M.D., Ph.D. [g], Christopher S. Monk, Ph.D. [a,c,e,h], Colter Mitchell, Ph.D. [e,f,*]

[a] Department of Psychology, University of Michigan, Ann Arbor, MI, USA
[b] College of Human Medicine, Michigan State University, Lansing, MI, USA
[c] Neuroscience Graduate Program, University of Michigan, Ann Arbor, MI, USA
[d] Wu Tsai Institute, Yale University, New Haven, CT, USA
[e] Survey Research Center of the Institute for Social Research, University of Michigan, Ann Arbor, MI, USA
[f] Population Studies Center of the Institute for Social Research, University of Michigan, Ann Arbor, MI, USA
[g] Department of Molecular Biology, Princeton University, Princeton, NJ, USA
[h] Michigan Neuroscience Institute, University of Michigan, Ann Arbor, MI, USA
[i] The Hilltop Institute, University of Maryland Baltimore County, Baltimore, MD, USA

ARTICLE INFO

Keywords:
Lead exposure
DNA methylation surrogate
Diffusion MRI
Development
Representative sample

ABSTRACT

Lead (Pb) is a potent neurotoxicant that can adversely affect health and cognitive function. This study used an epigenetic score of prenatal Pb exposure to assess associations between lead-related methylation and brain structure in adolescents. We studied 181 adolescents from the Study in Adolescent Neural Development (SAND), a subsample of the Future of Families and Child Wellbeing study, part of a longitudinal cohort with significant representation of marginalized youths. Potential prenatal Pb exposure was computed based on DNA methylation (DNAm) samples collected at ages 9 and 15. Diffusion MRI scans were collected at age 15 and used to produce white matter connectivity network metrics: small-world propensity, global efficiency, modularity, and transitivity. A higher prenatal DNAm Pb score at age 9 was linked to decreased structural global network efficiency in adolescence ($\beta = -0.26$, $p = .017$), transitivity ($\beta = -0.22$, $p = 0.04$), small-world propensity ($\beta = -0.246$, $p = .02$), as well as increased modularity ($\beta = 0.179$, $p = 0.03$) . Age 9 Pb methylation scores were particularly linked to network efficiency where the connections to and from the frontal orbital subnetwork were individually excluded. This suggests that this subnetwork is particularly important in the association between structural brain networks and surrogates of prenatal Pb exposure. Results indicate that a DNA methylation surrogate of prenatal lead exposure is associated with differences in white matter connectivity organization.

1. Introduction

Lead (Pb) is a pervasive and potent neurotoxicant that has lifelong consequences on health and cognitive function (Grönqvist et al., 2020; McFarland et al., 2022). Though blood Pb levels in children have declined in the United States, even small levels of Pb exposure are detrimental and can result in poor developmental outcomes associated with academic achievements, as well as behavioral problems, including impulsivity and delinquent behavior (Grönqvist et al., 2020; Winter & Sampson, 2017). Every year, 535,000 children in the United States have blood Pb levels at or above 3.5 μg/dL in blood, the Center for Disease Control (CDC) threshold for clinical concern (Grönqvist et al., 2020; McFarland et al., 2022; Wheeler & Brown, 2013). Importantly during pregnancy, prior maternal Pb exposure may be leached from bones during pregnancy and released into the bloodstream (Rísová, 2019). Susceptibility to lead exposure then begins during gestation as Pb freely passes through the placenta, making the Pb levels of the pregnant person of critical concern for preventing childhood lead exposure as well.

* Corresponding author at: Institute for Social Research, 426 Thompson St, Ann Arbor, MI 48104.
E-mail address: cmsm@umich.edu (C. Mitchell).

https://doi.org/10.1016/j.braen.2026.100013
Received 25 September 2025; Received in revised form 22 December 2025; Accepted 11 February 2026
Available online 13 February 2026
3050-5917/© 2026 Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

Exposure to maternal stress and toxicants, including Pb, during the prenatal period can alter functional and structural brain development (Lautarescu et al., 2020). In particular, Pb exposure impairs the development of the central nervous system at a cellular level by altering synaptogenesis and the release of neurotransmitters (Senut et al., 2012), influencing later neuropsychological development. These factors could also increase risk for neurodegenerative diseases (Llop et al., 2013; Senut et al., 2012; White et al., 2007), as well as reduced cognitive functioning (Naranjo et al., 2020).

Models of brain networks can offer more comprehensive representations of the brain by summarizing information across the entire connectome, rather than relying on the localized function or structure of a single brain region (Kragel et al., 2021; Taxali et al., 2021). Network neuroscience quantifies overall brain connectivity by describing properties of brain regions (nodes) and their connections (edges) (Dennis et al., 2013). These metrics can be used to summarize global and regional organization of white matter (e.g. integration, segregation, clustering, and small-world properties) through measures like global efficiency, modularity, transitivity, and small-world propensity. Examining the topological properties of the connectome are useful in predicting health and behavior, and organization of brain networks has been implicated in various developmental outcomes such as depression, Alzheimer's, and schizophrenia (Bassett & Bullmore, 2009). Taken together, these metrics work within a framework for understanding the complex architecture of brain networks by leveraging the strength of structural connections among brain regions to characterize how information is conveyed throughout the brain using white matter structures (Bullmore & Sporns, 2009; Hardi et al., 2023; Menon, 2011).

Changes in these metrics that describe global white matter organization demonstrate increased network integration and decreased network segregation across child development, and are associated with improved information processing efficiency through the selective strengthening and pruning of structural connections (Hagmann et al., 2010; Hardi et al., 2023; Rubinov & Sporns, 2010). Alterations to this pattern of global organization are associated with poor developmental outcomes. For example, decreased global efficiency throughout childhood, e.g. decreased network integration, has been associated with ADHD symptoms (Beare et al., 2017) and executive function impairments (Muñoz-Moreno et al., 2016). Moreover, the prenatal period is a key period for the establishment of these network properties, with small-world characteristics emerging as early as 20 postmenstrual weeks, and increasing global efficiency between 20 to 40 postmenstrual weeks (Song et al., 2017). Understanding the links between Pb exposure and the development of global white matter connectivity is key, as disruptions in structural organization can impact cognitive and socioemotional development (Griffiths et al., 2016; B. Wu et al., 2023; Yang et al., 2020).

One potential method to proxy Pb exposure is through the epigenetic signatures in DNA methylation (DNAm). DNAm regulates gene expression and is influenced by environmental inputs. Within the animal model literature there is strong evidence that Pb exposure throughout life has long-term causal effects on the DNA methylation (Faulk et al., 2013; Maloney et al., 2018; Senut et al., 2012, 2014; Varma et al., 2017). In humans, prenatal Pb exposure is associated with changes in DNAm that mediate the association between Pb and decreased cognitive outcomes in early childhood (Rygiel et al., 2021). Epigenome-wide association studies (EWAS) identify relationships between prenatal Pb exposure and methylation sites that correlate with multiple biological pathways including calcium transport, and signaling activities (Tung et al., 2022). The largest epigenome-wide exploration of the associations of prenatal lead in immune cells identified distinct DNAm alterations in umbilical cord blood associated with prenatal Pb exposure (as measured in maternal blood Pb levels) in a sample of 268 infant-mother pairs from a prospective pre-birth cohort (S. Wu et al., 2017).

Because of the long-term impact of Pb exposure on DNA methylation, it may be possible to estimate the associations of the exposure for an extended period. The weighted combinations of the methylation markers identified through EWAS, such as in S. Wu et al. (2017), has been used before to quantify exposure to physiological and psychological measurements (e.g. BMI and depression), as well as various toxicants such as pollution or maternal smoking during pregnancy (Bakulski et al., 2021; Clark et al., 2020; Hamilton et al., 2019; Richmond et al., 2018).

In the present study, we applied a polyepigenetic surrogate score trained on prenatal Pb exposure. This surrogate score of Pb exposure uses CpG sites and effect estimates identified from the largest available EWAS of prenatal Pb exposure conducted in Project Viva, where maternal blood Pb levels were measured during pregnancy and associated with umbilical cord-derived DNAm at birth (Wu et al., 2017). Applying these weights into the independent Future of Families and Child Wellbeing Study (FFCWS), which contains DNAm samples collected at later ages, allows for the estimation of a polyepigenetic index that reflects whether these DNAm patterns associated with prenatal Pb exposure measured in later developmental periods. Though this measure is trained on prenatal Pb exposure, there is little understanding if it is also associated with additional Pb exposure later in life, but will be referred to as a surrogate score of potential prenatal Pb exposure.

Because no prior studies have investigated associations between Pb exposure (or epigenetic markers of this exposure) and the organization of the structural connectome, we applied a polyepigenetic surrogate reflecting prenatal Pb-associated DNA methylation patterns to saliva-based DNAm samples collected at ages 9 and 15. Using data from the Study of Adolescent to Adult Neural Development (SAND), a subset of participants from the FFCWS population-based birth cohort, we examined whether higher surrogate scores of potential prenatal Pb exposure were associated with differences in global and regional structural connectivity. We aimed to (i) characterize the associations between the surrogate of prenatal Pb exposure derived from saliva DNAm at ages 9 and 15 with adolescent structural neural network architecture (age 15) using four metrics of overall brain connectivity small-world propensity, global efficiency, transitivity, and modularity, as well as (ii) to examine regional associations of the Pb methylation surrogate by exploring subnetwork specificity. We hypothesized that higher Pb methylation of the prenatal Pb exposure surrogate derived from saliva DNAm at ages 9 and 15 would be associated with differential patterns of structural network organization in adolescents.

2. Materials and methods

2.1. Study design and population

Briefly, participants were from the SAND cohort (Goetschius et al., 2019; Hardi et al., 2023; Hein et al., 2018), a subset of 237 participants recruited from FFCWS ($N = 4898$ cohort; Reichman et al., 2001). The SAND cohort consists of primarily Black non-Hispanic adolescents from large midwestern cities ($N = 237$; 76% Black, non-Hispanic). For participants who completed the broader FFCWS age 15 data collection, a subsample of families currently living in the three closest birth cohort cities (Detroit, Toledo, and Chicago) was invited to take part in additional data collection at the University of Michigan. Saliva DNA samples were taken at ages 9 and 15, and multimodal MRI was acquired at 15. Written and informed consent was obtained from the parents and adolescents provided verbal assent for their participation in SAND. All assessments and measures were approved by the Institutional Review Board of the University of Michigan (IRB: HUM00167754; HUM00074392).

At age 9, 38 participants did not provide a saliva sample, and at age 15, 12 participants did not provide a saliva sample. The final analytic sample consisted of 181 participants with complete MRI, DNAm, and covariate data. Among the final sample participants were primarily African American ($n_{\text{African American}} = 137$, 78%), female ($n = 99$, 55%), and born in Detroit ($n_{\text{Detroit}} = 119$, 66%). Participants were excluded

based on their completion of the MRI scan (i.e. refusal of MRI, exceeded weight limit, medical restrictions, braces or other metal in the body, pregnancy, Autism Spectrum Disorder diagnosis, or not completing the dMRI scan), and imaging artifact (identified in preprocessed structural or tractography during quality control, or extreme outliers in network measure). A full breakdown of descriptive statistics for the full FFCWS and final sample are available in Table 1, and see Supplemental Table S1 for total counts for each exclusion criteria.

2.2. Measures

2.2.1. DNA methylation collection and processing

Approximately 500 ng of genomic DNA (quantified using the Quant-iT Picogreen ds DNA Assay Kit) was subjected to bisulfite conversion using the EZ-96 DNA Methylation Kit (Zymo Research) and analyzed with the Illumina Infinium Human Methylation 450 K (450 K) according to the manufacturer's protocol. Age 9 and 15 DNA samples were run at the same time to minimize technical variation. Otherwise, paired samples were randomized. The red and green image pairs were read into R (R Core Team, 2021) and QC of the methylation data was initially performed with EWAStools (Heiss & Just, 2019). Probes were removed if the detection value was greater than 0.01. Probes were also removed if the number of methylated or unmethylated bead count was fewer than four. Probes were also removed if they were identified by the ENmix function QCinfo using the default parameters (Xu et al., 2016). Samples were removed if they had outlier methylation or bisulfite conversion values, as identified by the ENmix QC function or if the sex predicted from the methylation data differed from the recorded sex. The ENmix preprocessENmix and rcp functions were used to normalize dye bias, apply background correction and adjust for probe-type bias (Niu et al., 2016; Xu et al., 2016). Relative cell proportions of epithelial cells, fibroblasts, and immune cells were calculated using EpiDISH, which infers the proportion of known cell-types in a sample.

Table 1
Sample comparisons across full FFCWS sample (N=4,898) and SAND sample (N=181).

Variable	FFCWS N = 4,898 M (SD)	SAND N = 181 M (SD)
Age at wave 5	9.3 (0.31)	9.2 (0.32)
Age at wave 6	15.6 (0.8)	15.7 (0.9)
Income to needs ratio at baseline	2.22 (2.41)	1.98 (2.27)
Income to needs ratio at 15	2.40 (2.66)	1.95 (2.19)
White matter connectivity metrics		
Small-world propensity	-	0.866 (0.025)
Global efficiency	-	0.063 (0.01)
Modularity	-	0.586 (0.016)
Transitivity	-	0.0031 (0.0005)
	n (%)	n (%)
Sex assigned at birth		
Male	2,568 (52%)	82 (45%)
Female	2,329 (48%)	99 (55%)
Race/ethnicity		
African American	1,601 (49%)	137 (78%)
Hispanic	813 (25%)	12 (6.8%)
European American	590 (18%)	20 (11%)
Other	261 (8%)	7 (4%)
Birth city		
Other	4,314 (88%)	5 (2.8%)
Detroit	327 (6.7%)	119 (66%)
Chicago	155 (3.2%)	26 (14%)
Toledo	101 (2.1%)	31 (17%)
Mother's education level at baseline		
Less than high school	1,699 (35%)	63 (35%)
High school or equiv.	1,480 (30%)	51 (28%)
Some college, or tech school	1,189 (24%)	46 (25%)
College or graduate	524 (11%)	21 (12%)
Low birth weight		
No	4275 (90%)	148 (83%)
Yes	484 (10%)	30 (17%)

2.2.2. DNAm surrogate of prenatal Pb exposure

Prenatal polyepigenetic surrogates of potential Pb exposure were computed using statistically significant CpG sites identified in S. Wu et al. (2017) after false discovery rate (FDR) correction (Benjamini & Hochberg, 1995). Methylation beta values were converted into M values. After mean centering and standardizing the selected CpG sites, these scores were calculated by taking the linear combination of the m-values for all fdr-significant CpG sites identified by S. Wu et al. (2017) multiplied by that CpG site's reported coefficient in the epigenome-wide association study (EWAS). These coefficients were derived from the association strengths using the results of the EWAS of DNAm in umbilical cord blood associated with maternal red blood cell Pb concentrations for all newborns. DNAm surrogates of potential prenatal Pb exposure were residualized for immune cell proportion, fibroblast proportion, batch, and mother's smoking status at birth.

2.2.3. MRI image acquisition and preprocessing

At age 15, participants completed a neuroimaging protocol including assessment of diffusion MRI (dMRI). Images were acquired using a 3T GE Discovery MR750 with 8-channel head coil. T1-weighted gradient echo images were captured (TR=12 ms, TE=5 ms, TI=500 ms, flip angle=15°, field of view=26 cm, slice thickness=1.44 mm, 256×192 matrix, 110 slices). dMRI data were acquired using spin-echo EPI diffusion sequence using repetition time of 7250 ms, minimum echo time, 128×128 acquisition matrix, field of view of 22 cm, 3 mm no-gap thick slices with 40 slices acquired using alternating-increasing order, bvalue of 1000s/mm2, 64 non-linear directions, five b = 0s/mm2 T2 images (b0) images were acquired. dMRI images were first inspected visually for quality. Slices with an average intensity > 4 standard deviations or more were marked as outliers and replaced with predicted models (Andersson et al., 2016). Participants were also excluded if more than 5% of slices were replaced and images for the 10 participants with the highest proportion of replaced slices went through further visual inspection. This data was used and also described in previous publications (Calabrese et al., 2022; Goetschius et al., 2019, 2020; Hardi et al., 2022, 2023; Hein et al., 2018).

2.2.4. dMRI data processing

dMRI data processing was previously reported in Hardi et al., (2023), but is described here. Data from dMRI were processed using the MRtrix pipeline (V2; (Tournier et al., 2007). MRtrix utilized a novel tensor-fitting method called the Constrained Spherical Deconvolution (CSD; Farquharson et al., 2013; Tournier et al., 2004, 2007) that has been found to outperform DTI or other alternative methods in crossing fibers regions, which are poorly accounted for using conventional deterministic methods. Probabilistic tractography was used to estimate ten million streamlines, which were then used to estimate structural connectivity using the anatomical AAL2 atlas. This created a 94×94 connectome matrix representing the count of streamlines or structural connectivity between each distinct node or brain region. Additionally, these nodes were also subdivided into 8 subnetworks based on the AAL2 parcellation (Rolls et al., 2015). See Supplemental Table S2 for the full list of coordinates for nodes in each subnetwork from the AAL2 parcellation adapted from Hardi et al. (2023), and Supplemental Section 1 for descriptions of formulas used to calculate white matter connectivity metrics.

The resulting white matter connectome data were subsequently processed as weighted, undirected, and unthresholded graphs using the Brain Connectivity Toolbox in MATLAB to generate global and local metrics of network architecture: global efficiency, modularity, transitivity, and small-world propensity (Rubinov & Sporns, 2010). Global efficiency, a measure of network integration, quantifies the average inverse shortest path length in the network, reflecting the ease of information transfer between distant brain regions (Fagiolo, 2007; Latora

& Marchiori, 2001; Onnela et al., 2005; Rubinov & Sporns, 2010; Y. Wang et al., 2017). Modularity, a measure of network segregation, assesses the degree to which the network can be divided into distinct communities (Newman, 2006; Reichardt & Bornholdt, 2006). Transitivity, or clustering coefficient, measures the likelihood that two neighbors of a node are also connected, reflecting the prevalence of tightly-knit clusters within the network (Onnela et al., 2005). Additionally, the balance between network integration and segregation can also be summarized using measures such as small-world propensity, that indicates the extent to which the network is characterized by small-world properties (Muldoon et al., 2016). It indexes how nodes in close proximity within a network are densely connected, but physically distant nodes are still closely connected through neighbors, e.g. each node has no more than "six degrees of freedom" to other nodes within the network (Watts & Strogatz, 1998 ;Latora & Marchiori, 2001; Muldoon et al., 2016). As an exploratory analysis to determine the importance of specific regions in network computation, a targeted attack approach adapted from (Siugzdaite et al., 2020) was taken. Using this approach, subnetworks were individually dropped by setting the strength of white matter pathways coming in and leaving the nodes within a region to 0 before recalculating global efficiency. This process was repeated for each subnetwork, resulting in 8 separate network metrics for all participants, each characterizing the network without one specific subnetwork. Examining the difference in network characteristics after excluding a subnetwork allows us to identify the contribution of specific subnetwork to overall network organization.

2.2.5. Other covariate measures

Participant race/ethnicity, age, sex (assigned at birth), birth city (categorical variable binned into 7 categories), mother's baseline education (ordinal variable binned into 4 categories), low birth weight (categorical variable binned into 2 categories), and income-to-needs ratio of primary caregiver at birth and age 15 were included as covariates in the analysis. Race was included as a socially constructed category to account for unequal exposure to racism and discrimination, participant age at the time of DNAm data collection is included to adjust for differences in white matter connectivity due to differences in age (Hagmann et al., 2010), sex was included to control for potential sex dependent differences in white matter structural organization (Ingalhalikar et al., 2014), while birth city was included to account for sampling differences by geographical location. Race, mother's education at birth, and income to needs ratio were included to account for potential correlations to childhood or adolescent experiences that relate to Pb exposure (Teye et al., 2021).

2.3. Statistical analyses

We applied linear models to determine associations between the surrogate of potential prenatal Pb exposure scores at 9 or 15 and white matter connectivity network metrics, while accounting for demographic covariates listed in the previous section. Tests within each timepoint were corrected for multiple comparisons using false discovery rate correction (Benjamini & Hochberg, 1995). Distributions of key predictor and outcome variables are available in Supplemental Figure S1.

3. Results

3.1. Reliability of DNAm surrogates of potential prenatal Pb exposure

Correlation tests were run on the DNAm surrogates of potential prenatal Pb exposure at ages 9 and 15, which demonstrated a moderate correlation between DNAm surrogates between ages 9 and 15, suggesting stability in scores between the two timepoints ($t(136) = 9.057$, $r = .61$, $p < .001$). Additionally, for 45 randomly selected samples, we reran their sample in another batch in order to be able to estimate the between batch reliability of methylation data and scores. For the age 9 methylation data, the DNAm surrogate of potential prenatal Pb exposure had an ICC = 0.69, and the age 15 DNAm surrogate had a slightly lower reliability (ICC = 0.64), suggesting the DNAm surrogate at ages 9 and 15 demonstrate moderate reliability. When we looked at 9 technical replicates (samples rerun on the same plate), scores were more reliable (ICC at age 9 = 0.88 and age 15 = 0.86), suggesting the technical replicates demonstrate good reliability (Koo & Li, 2016).

3.2. Associations between DNAm surrogate of potential prenatal Pb exposure and white matter network connectivity metrics

DNAm surrogate of potential prenatal Pb at age 9 showed significant first-order negative associations with global efficiency (network integration; $\beta = -0.20$, SE = 0.08, $p = 0.02$), transitivity (clustering; $\beta = -0.17$, SE = 0.08, $p = 0.03$), and small-world propensity (network integration and segregation balance; $\beta = -0.22$, SE = 0.08, $p = 0.02$), as well as a positive association with modularity (network segregation; $\beta = 0.20$, SE = 0.08, $p = 0.02$). These associations also persisted after adjusting for covariates (global efficiency: $\beta = -0.25$, SE = 0.09, $p = 0.019$; transitivity: $\beta = -0.232$, SE = 0.09, $p = 0.019$ small-world propensity: $\beta = -0.21$, SE = 0.09, $p = 0.025$; and modularity: $\beta = 0.2$, SE = 0.09, $p = 0.02$). All associations remained significant after controlling for covariates and multiple comparisons. Full table of regression estimates between DNAm surrogate of potential prenatal Pb at 9 with small world propensity, global efficiency, transitivity, and modularity at age 15 are available in supplemental table S3. This suggests that higher DNAm surrogate scores of potential prenatal Pb at age 9 were associated with decreased network integration, decreased clustering, increased network segregation. Moreover these scores are also associated with metrics of higher order organization that demonstrates an altered balance between network integration and segregation.

There were no associations between DNAm surrogate scores of potential prenatal Pb at 15 and any of the four white matter metrics. Scatterplots of the associaitons between the surrogate scores and white matter metrics are available in Fig. 1. Full table of regression estimates for associations between DNAm scores of potential prenatal Pb at 15 and with small-world propensity, global efficiency, transitivity, and modularity at 15 are available in supplemental table S4.

3.3. Sensitivity analyses

3.3.1. Bias in age 15 analyses related to participant exclusion

Because of the lack of association between the DNAm surrogate of potential prenatal Pb exposure at 15 and global efficiency may be driven by participants who provided methylation data at 15 but not at 9. To test this, we then ran additional analyses between the DNAm score at age 15 and all four white matter metrics (small world propensity, global efficiency, modularity, and transitivity) at age 15 using a subset of participants who had also provided age 9 methylation data. There were no significant associations between DNAm surrogate scores of potential prenatal Pb exposure and any brain metric age 15 in this sub-sample, suggesting that the lack of associations between Pb and brain metrics were not driven by participants who did not provide methylation data at age 9. Results for these analyses are available in supplemental table S5.

3.3.2. Additional associations between DNAm surrogate scores and confounding variables

Additional analyses were conducted to examine associations between the DNAm surrogate of potential prenatal Pb and developmental factors, including pubertal status and age. Associations between the scores and pubertal status were conducted since puberty is associated with widespread epigenetic remodeling, but did not show any significant association with the DNAm surrogate of potential prenatal Pb exposure at age 15 ($t(167) = .085$, $r = 0.006$, $p = 0.93$). This suggests that puberty-related biological changes are unlikely to explain the lack of association at age 15. Other analyses were conducted to examine if

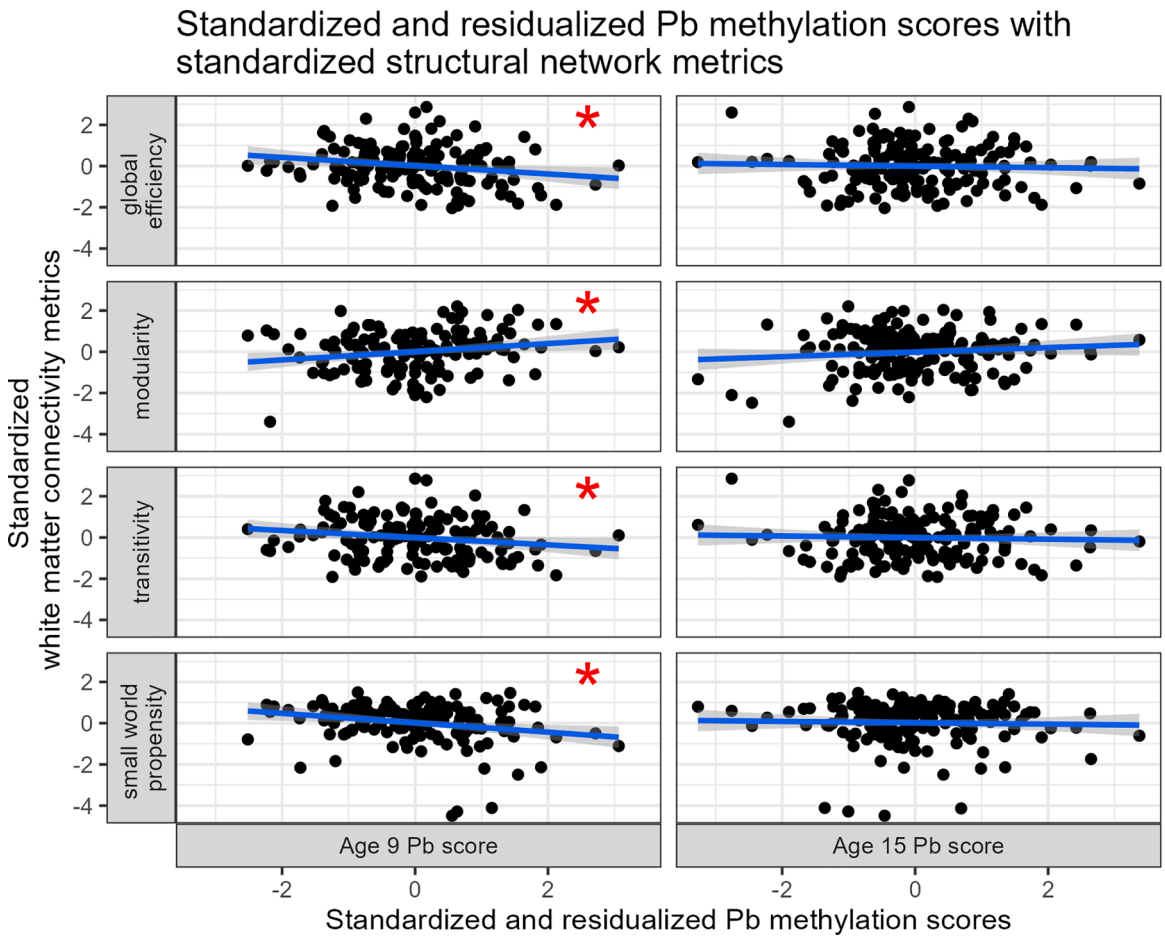

Fig. 1. Associations between standardized white matter network connectivity metrics and standardized surrogate of prenatal Pb exposure scores at ages 9 and 15.

Blue lines represent regression lines between DNAm surrogate scores of prenatal lead at age 9 (left) and age 15 (right) with each measure of white matter connectivity (global efficiency, modularity, transitivity, small-world propensity). DNAm surrogate scores of prenatal lead at age 9 predicted global efficiency, modularity, transitivity, and small-world propensity at age 15. Red asterisks denote significant associations FDR-corrected $p < .05$.

age at each timepoint was associated with the DNAm surrogate of prenatal Pb at each timepoint. Participant age at FFCWS wave 5 (age 9), was not correlated with the DNAm surrogate at age 9 ($t(141) = -0.56$, $r = -0.05$, $p = 0.5$), but participant age at FFCWS wave 6 (age 15) was significantly associated with the DNAm score ($t(167) = 2.21$, $r = 0.17$, $p = 0.03$). This may be due to selection bias, as a subset of participants at the age 15 wave were collected at a much later time. The removal of these 23 participants removed this age association ($t(144) = 1.22$, $r = 0.1$, $p = 0.22$), suggesting the association between DNAm scores at 15 and participant age are driven by this subset of participants collected at a later time. Despite the significant association between age at wave 6 and the DNAm score, the inclusion of age at wave 6 as a covariate in the analyses was not predictive of white matter connectivity at 15. This suggests that although age contributes small differences in DNAm scores at adolescence, adjusting for this variable does not alter associations with brain outcomes.

3.3.3. *Exploratory analysis examining subnetwork effects on global efficiency*

Following the targeted attack approach from Siugzdaite et al., (2020), recomputation of network metrics was applied to examine the contribution of each subnetwork to global efficiency. Subnetworks (e.g., DMN) were individually dropped by setting the strength of white matter pathways entering and leaving the nodes within a subnetwork to 0 before recalculating global efficiency. This process was subsequently repeated, each time excluding one particular subnetwork before efficiency was recalculated. As the hubs were removed, only the global efficiency of connectomes where the frontal orbital ($\beta = 0.06$, SE $= 0.08$, $p = 0.5$) subnetwork was removed showed a decrease in the regression coefficient with the DNAm surrogate of potential prenatal Pb exposure after FDR-correction. Individually, the exclusion of connectivity involving the frontal orbital subnetwork attenuated the association between DNAm scores of prenatal Pb at age 9 and global efficiency, suggesting that the connectivity of these regions may be most important in explaining the association between global efficiency and prenatal Pb methylation. Standardized coefficients and scatterplots of the associations are shown in Fig. 2 and see Supplemental Table S6 for the full table of regression coefficients.

3.3.4. *Examining changes in variability between timepoints*

To examine whether the methylation signature from the DNAm surrogate becomes less biologically informative over time, we tested for changes in variability in the score through a Pitman-Morgan test of variance on residualized but not standardized versions of the scores across both timepoints. Greater variance at later ages may reflect increased noise of the original prenatal exposure-related signal. There were significant differences between ages 9 and 15 ($t(136) = 2.47$, $p = 0.015$), suggesting that the DNAm surrogate of prenatal Pb showed higher variability at age 15 than age 9. This pattern is consistent with a decay in meaningful signal between the two timepoints. Violin plots depicting the change in distribution between ages 9 and 15 are shown in Fig. 3.

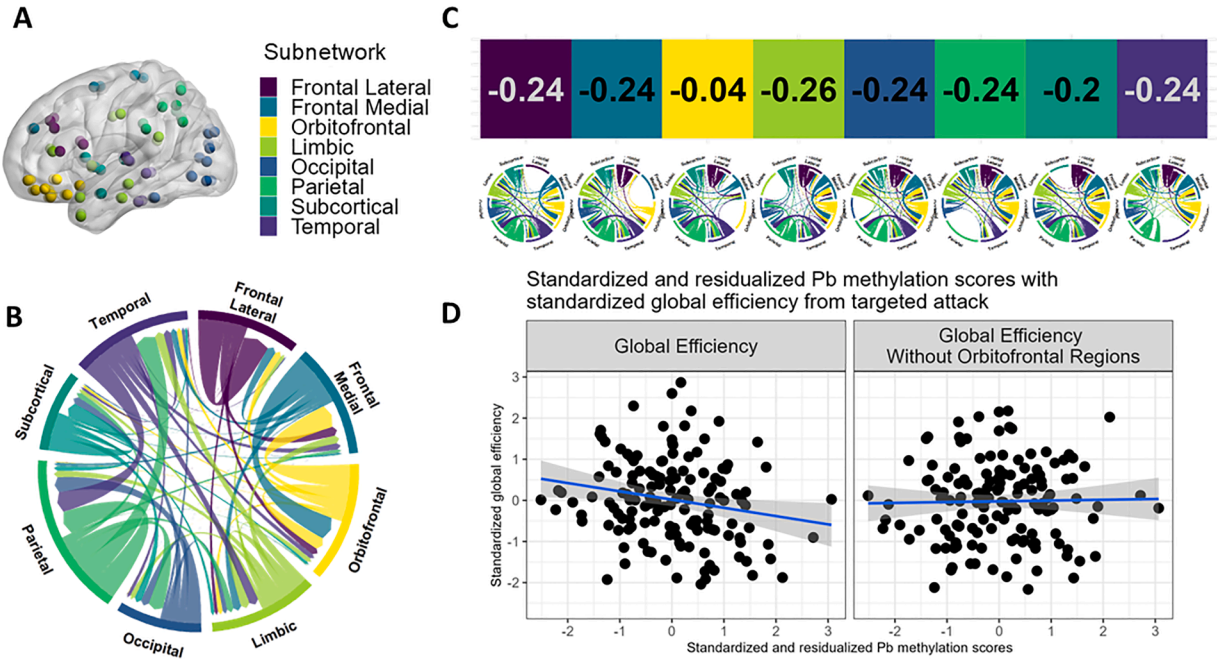

Fig. 2. Identification of subnetwork-specific effects of Pb methylation on white matter organization.
A. Visual representation of the nodes within each subnetwork (frontal lateral, frontal medial, orbitofrontal, limbic, occipital, parietal, subcortical, and temporal)
B: Circular plot of white matter structural connectivity from one sample participant. Each colored arc on the circle represents structural connectivity originating from each subnetwork and the thickness represents the connectivity strength.
C. Associations between prenatal Pb score at 9 and recalculated global efficiency from the targeted attack analysis. Each color represents the brain region that was dropped before global efficiency was recalculated. Each number represents the standardized regression coefficient between one recalculated global efficiency and the DNAm surrogate scores of prenatal lead at age 9.
D. Illustration of networks excluding a particular region. Regions were dropped by setting the strength of white matter pathways coming in and leaving the nodes within a region to 0 before recalculating global efficiency. Axes show sample circular plots where the respective region's connectivity is set to 0.

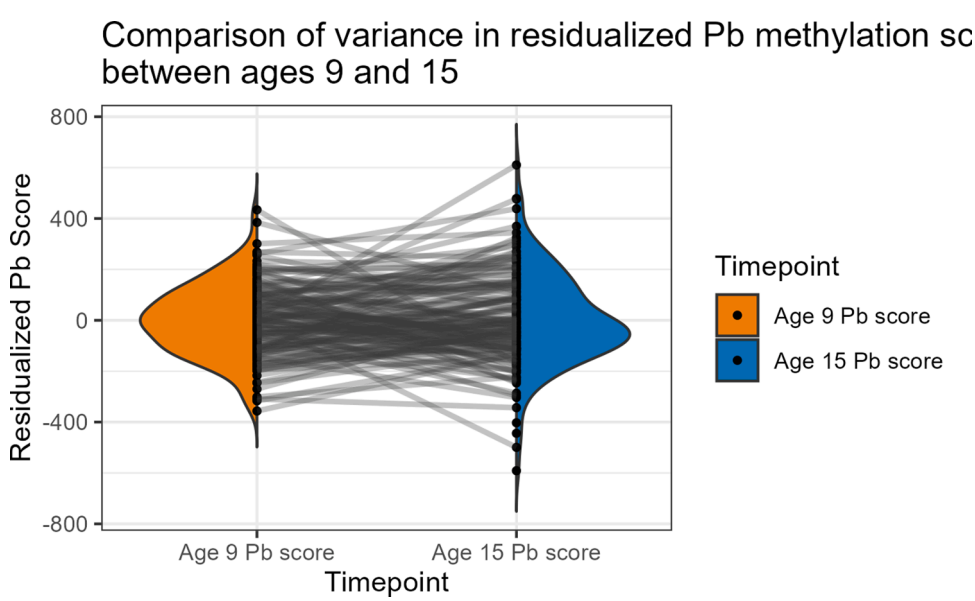

Fig. 3. Testing the change in variance between ages 9 and 15.
Violin plots of residualized Pb scores at age 9 (left) and age 15 (right). Black lines show change in score for individual participants between two timepoints.

4. Discussion

This study examined the relationship between DNAm scores associated with potential prenatal Pb exposure measured at two developmental periods (childhood and mid-adolescence) with structural network architecture metrics at age 15. We found that higher potential prenatal Pb scores at age 9 were associated with less integrated, less clustered, and more segregated white matter brain networks, as well as a reduced balance between network integration and segregation. Moreover, further analysis found the frontal orbital subnetwork to be an important region for the association between global efficiency and DNAm scores of potential prenatal Pb exposure.

Prenatal Pb exposure is known to cause disruptions in neurotransmitter and synaptic function, which may disrupt the efficiency of

information transfer across white matter tracts (Ma et al., 1999; White et al., 2007). Highly efficient networks are better at integrating information from different brain regions, enabling coordinated and integrated processing of stimuli, thoughts, and actions (Betzel & Bassett, 2018). Conversely, cognitive and developmental disorders, such as ADHD, are associated with breakdowns in this information flow (Cao et al., 2016). Associations with a measure of network integration may occur because Pb is known to impact long-range and cross hemispheric structural connectivity, which support complex brain dynamics by increasing network input and output diversity (Betzel & Bassett, 2018; Thomason et al., 2019). The development of long-range white matter fibers during the third trimester is associated with increase in global efficiency within the structural connectome (Song et al., 2017). Moreover, the disruption to development of those white matter fibers also have implications for measures of network segregation. Long-range white matter fibers are known to terminate between specialized modules (Taylor et al., 2017). Therefore, fewer long-range connections may decrease integration parameters while increasing segregation parameters (Frey et al., 2021; Thomason et al., 2019).

Decreased network integration and increased segregation may point towards disrupted network organization resulting in slower interactions between brain regions and decreased ability to integrate information (Li et al., 2012). This is reflected in the negative association between DNAm scores of prenatal Pb and small-world propensity in the present study, as small-world propensity summarizes across information about the balance between network integration and segregation (Liao et al., 2017). Because small-world properties of brain networks emerge at birth (Chirumamilla et al., 2023), reduction in small-world propensity may reflect neurodevelopmental delays associated with early Pb exposure. Since the DNAm surrogate used in present study is based on methylation patterns associated with maternal Pb levels at the time of the child's birth, these patterns most likely reflect maternal blood levels from around the third trimester. However, the precise developmental timing cannot be determined in the available data, nor did the previous study establishing this measure of Pb-related epigenetic patterns test for the developmental specificity of this exposure.

Findings from the targeted attack approach revealed specific regions of interest, specifically, the frontal orbital subnetwork that were important for maintaining the association between white matter network organization and the DNAm surrogate. The orbitofrontal cortex is a region implicated in a number of higher order executive functions, such as inhibitory control (e.g. impulse control; Friedman & Robbins, 2022) and affective regulation (Bechara et al., 2000), which are also deficits associated with Pb exposure (Grönqvist et al., 2020; Winter & Sampson, 2017). These findings are in line with previous research which identified associations between childhood lead exposure and decreased prefrontal gray matter volumes in adulthood, which may drive cognitive and behavioral problems seen in Pb exposure (Cecil et al., 2008). Other research in adults with high lead exposure suggests that the frontal lobes are particularly sensitive to blood and bone lead levels using proton magnetic resonance spectroscopy (Hsieh et al., 2009). Research on fetal white matter development suggests that the third trimester is associated with the development of connective fibers including fibers that connect the frontal lobes to more distal areas like the occipital lobe (Song et al., 2017; Thomason, 2020). Taken together with the findings of the targeted attack approach, Pb exposure may interfere with development of early white matter organization with particular implications on cognitive functioning through the orbitofrontal cortex. These results corroborate findings on global and local effects of Pb exposure on brain structure and function, and may provide another mechanism in which Pb influences cognitive function through brain anatomy.

There were no associations between the DNAm surrogate of potential prenatal Pb at age 15 and any brain metrics at 15. Lack of an association between may reflect a continual decay in Pb methylation signal since birth to a point that signal cannot be well quantified by age 15, possibly as a result of in utero and early life Pb exposures being deposited in bone rather from erythrocytes (Rísová, 2019; Wani et al., 2015). Current research suggests that stability of DNAm probes are dynamic across time and are further modified by the presence of different exposures across the lifecourse (Apsley et al., 2023; Danielewski et al., 2024). Moreover, similar patterns have been observed in other DNAm scores, such as with DNAm scores predicting age, which identified DNAm probes that were previously associated with age, demonstrating inconsistent direction of change across a 15-year interval (Refn et al., 2023). This is also consistent with findings from other DNAm scores calculated within FFCWS, such as with prenatal maternal smoking exposure, which found that DNAm signal associated with it degrades over time (Blostein et al., 2022). This suggests that as the exposure becomes further removed in time, subsequent environmental influences may further erode the signal, which may drive the lack of associations between DNAm scores of potential prenatal Pb exposure at age 15 and white matter connectivity and the moderate correlations in scores between timepoints. These moderate correlations between DNAm scores at each timepoint are also comparable to the reliability of other epigenetic measures, such as that of the Paediatric ($r = 0.49$) or GrimAge ($r = 0.49$) (Blostein et al., 2022). Moreover, DNAm at age 15 is much farther removed from the prenatal period, suggesting that more proximal measures of Pb should be explored at this time period. It is also possible that major developmental changes that occur during adolescence, including major remodeling of the prefrontal cortex, may smooth altered trajectories of brain development from prenatal exposures. That is, the large brain changes that occur in adolescence could essentially wipe away some of the effects of early exposure, a hypothesis that would have important implications for resilience and one that would require more extensive longitudinal data of both brain organization and Pb exposure.

The DNAm surrogate was used as an indirect measure of potential prenatal Pb exposure for this analysis, showcasing how epigenetic changes may serve as biomarkers of early environmental influences, including Pb exposure. Although studies such as Sha et al., (2023) have identified genetic loci associated with variability in white matter fiber tracts, and Mansel et al., (2019) and Senut et al., (2014) have linked Pb exposure to changes in methylation status in cell culture, direct evidence for epigenetic modifications as a toxic mechanism of Pb in the brain is still emerging. Nonetheless, these studies suggest that early exposure to Pb may disrupt normative brain development, including the formation of white matter connectivity, during a key period of white matter development (Lebel & Deoni, 2018).

Though the current study has strengths including the combination of novel measures of epigenetic markers associated with prenatal prenatal Pb exposure and metrics of white matter organization, there are some limitations. First, there may be a decay in methylation signal as there was a considerable time between when methylation was sampled and birth. Also, the extent to which additional Pb exposures between birth and age 9 influence the same methylation signature may impact these scores, as Pb exposure later in life may be associated with both similar and different DNAm markers. To better understand these trajectories would require continuous, prospective measurement of Pb exposure and DNAm to observe how associations between DNAm markers and Pb exposure change over time and in relation to brain organization. Current understanding of how DNAm surrogate scores of time-sensitive exposures (e.g. prenatal Pb, or prenatal smoking) are limited, as few studies have examined the consistency of scores over time. Second, the subsample used in this study is limited to births in large midwestern cities of the United States, which may limit the generalizability of associations to other populations, though these samples are representative of the cities they are sampled from. Finally, the DNAm surrogate of potential Pb exposure was created as a weighted linear combination of FDR-significant sites reported by the EWAS run by Wu et al. (2017). The method of creating surrogates used in this study may potentially double count effects of Pb on methylation for clustered CpG sites, however, alternative methods require that the algorithm be trained within the original sample.

5. Conclusion

In summary, the present study found higher potential prenatal Pb exposure methylation scores at age 9 were associated with lower global efficiency, transitivity, and small-world propensity at 15, but higher modularity at 15.

CRediT authorship contribution statement

Ryan Tung: Writing – original draft, Visualization, Formal analysis, Conceptualization. **Carly O'Neill:** Writing – review & editing, Data curation, Conceptualization. **Trey Smith:** Writing – review & editing, Software, Data curation. **Felicia A. Hardi:** Writing – review & editing, Supervision. **Leigh G. Goetschius:** Writing – review & editing, Data curation. **Edward D. Huntley:** Writing – review & editing, Conceptualization. **Helen C.S. Meier:** Writing – review & editing. **Luke W. Hyde:** Writing – review & editing, Project administration, Funding acquisition. **Dan Notterman:** Writing – review & editing, Data curation. **Christopher S. Monk:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Colter Mitchell:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We would like to acknowledge the past work of the Future of Families and Child Wellbeing Study, the participants and their families for sharing their experiences with us, and the project staff who make this research possible.

This work was supported by the National Institutes of Health [R01 MH103761 (Monk), R01 HD076592 (Notterman), R01 MD011716 (Mitchell), R01 HD036916 (Edin & Waldfogel), R01 MH121079 (Hyde, Mitchell, Monk)] and the Eunice Kennedy Shriver National Institute of Child Health and Human Development [T32 HD007109 (McLoyd, Monk)].

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.braen.2026.100013.

Data availability

Diffusion data from SAND will be openly available at https://nda.nih.gov/edit_collection.html?id=2106. Access to FFCWS data can be granted via a contract data use license to faculty and research personnel at institutions with an IRB or human subjects review committee registered with the US OHRP or NIH, after obtaining IRB approval for their research. To apply, contact ude.notecnirp@atadff with the required documents listed at https://ffcws.princeton.edu/restricted. For further questions, reach out to ude.notecnirp@atadff.

References

Andersson, J. L. R., Graham, M. S., Zsoldos, E., & Sotiropoulos, S. N. (2016). Incorporating outlier detection and replacement into a non-parametric framework for movement and distortion correction of diffusion MR images. *NeuroImage, 141*, 556–572. https://doi.org/10.1016/j.neuroimage.2016.06.058

Apsley, A. T., Ye, Q., Etzel, L., Wolf, S., Hastings, W. J., Mattern, B. C., … Shalev, I. (2023). Biological stability of DNA methylation measurements over varying intervals of time and in the presence of acute stress. *Epigenetics, 18*(1), 2230686. https://doi.org/10.1080/15592294.2023.2230686

Bakulski, K. M., Fisher, J. D., Dou, J. F., Gard, A., Schneper, L., Notterman, D. A., Ware, E. B., & Mitchell, C. (2021). Prenatal particulate matter exposure is associated with saliva DNA methylation at age 15: Applying cumulative DNA methylation scores as an exposure biomarker. *Toxics, 9*(10). https://doi.org/10.3390/toxics9100262. Article 10.

Bassett, D. S., & Bullmore, E. T. (2009). Human brain networks in health and disease. *Current Opinion in Neurology, 22*(4), 340–347. https://doi.org/10.1097/WCO.0b013e32832d93dd

Beare, R., Adamson, C., Bellgrove, M. A., Vilgis, V., Vance, A., Seal, M. L., & Silk, T. J. (2017). Altered structural connectivity in ADHD: A network based analysis. *Brain Imaging and Behavior, 11*(3), 846–858. https://doi.org/10.1007/s11682-016-9559-9

Bechara, A., Damasio, H., & Damasio, A. R. (2000). Emotion, decision making and the orbitofrontal cortex. *Cerebral Cortex, 10*(3), 295–307. https://doi.org/10.1093/cercor/10.3.295

Benjamini, Y., & Hochberg, Y. (1995). Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society. Series B (Methodological), 57*(1), 289–300.

Betzel, R. F., & Bassett, D. S. (2018). Specificity and robustness of long-distance connections in weighted, interareal connectomes. *Proceedings of the National Academy of Sciences, 115*(21), E4880–E4889. https://doi.org/10.1073/pnas.1720186115

Blostein, F. A., Fisher, J., Dou, J., Schneper, L., Ware, E. B., Notterman, D. A., … Bakulski, K. M. (2022). Polymethylation scores for prenatal maternal smoke exposure persist until age 15 and are detected in saliva in the Fragile Families and Child Wellbeing cohort. *Epigenetics, 17*(13), 2223–2240. https://doi.org/10.1080/15592294.2022.2112815

Bullmore, E., & Sporns, O. (2009). Complex brain networks: Graph theoretical analysis of structural and functional systems. *Nature Reviews Neuroscience, 10*(3), 186–198. https://doi.org/10.1038/nrn2575

Calabrese, J. R., Goetschius, L. G., Murray, L., Kaplan, M. R., Lopez-Duran, N., Mitchell, C., … Monk, C. S. (2022). Mapping frontostriatal white matter tracts and their association with reward-related ventral striatum activation in adolescence. *Brain Research, 1780*, 147803. https://doi.org/10.1016/j.brainres.2022.147803

Cao, M., Huang, H., Peng, Y., Dong, Q., & He, Y. (2016). Toward developmental connectomics of the human brain. *Frontiers in Neuroanatomy, 10*. https://www.frontiersin.org/articles/10.3389/fnana.2016.00025.

Cecil, K. M., Brubaker, C. J., Adler, C. M., Dietrich, K. N., Altaye, M., Egelhoff, J. C., Wessel, S., Elangovan, I., Hornung, R., Jarvis, K., & Lanphear, B. P. (2008). Decreased brain volume in adults with childhood lead exposure. *PLOS Medicine, 5*(5), e112. https://doi.org/10.1371/journal.pmed.0050112

Chirumamilla, V. C., Hitchings, L., Mulkey, S. B., Anwar, T., Baker, R., Larry Maxwell, G., … Govindan, R. B. (2023). Functional brain network properties of healthy full-term newborns quantified by scalp and source-reconstructed EEG. *Clinical Neurophysiology, 147*, 72–80. https://doi.org/10.1016/j.clinph.2023.01.005

Clark, S. L., Hattab, M. W., Chan, R. F., Shabalin, A. A., Han, L. K. M., Zhao, M., Smit, J. H., Jansen, R., Milaneschi, Y., Xie, L. Y., van Grootheest, G., Penninx, B. W. J. H., Aberg, K. A., & van den Oord, E. J. C. G. (2020). A methylation study of long-term depression risk. *Molecular Psychiatry, 25*(6). https://doi.org/10.1038/s41380-019-0516-z. Article 6.

Danielewski, M., Walkowiak, J., Wielgus, K., & Nowak, J. K. (2024). Stability of blood DNA methylation across two timepoints in three cohorts. *Biomedicines, 12*(11), 2557. https://doi.org/10.3390/biomedicines12112557

Dennis, E. L., Jahanshad, N., McMahon, K. L., de Zubicaray, G. I., Martin, N. G., Hickie, I. B., Toga, A. W., Wright, M. J., & Thompson, P. M. (2013). Development of brain structural connectivity between ages 12 and 30: A 4-tesla diffusion imaging study in 439 adolescents and adults. *NeuroImage, 64*, 671–684. https://doi.org/10.1016/j.neuroimage.2012.09.004

Fagiolo, G. (2007). Clustering in complex directed networks. *Physical Review E, 76*(2), Article 026107. https://doi.org/10.1103/PhysRevE.76.026107

Farquharson, S., Tournier, J.-D., Calamante, F., Fabinyi, G., Schneider-Kolsky, M., Jackson, G. D., & Connelly, A. (2013). White matter fiber tractography: Why we need to move beyond DTI: Clinical article. *Journal of Neurosurgery, 118*(6), 1367–1377. https://doi.org/10.3171/2013.2.JNS121294

Faulk, C., Barks, A., Liu, K., Goodrich, J. M., & Dolinoy, D. C. (2013). Early-life lead exposure results in dose- and sex-specific effects on weight and epigenetic gene regulation in weanling mice. *Epigenomics, 5*(5), 487–500. https://doi.org/10.2217/epi.13.49

Frey, B. M., Petersen, M., Schlemm, E., Mayer, C., Hanning, U., Engelke, K., Fiehler, J., Borof, K., Jagodzinski, A., Gerloff, C., Thomalla, G., & Cheng, B. (2021). White matter integrity and structural brain network topology in cerebral small vessel disease: The Hamburg city health study. *Human Brain Mapping, 42*(5), 1406–1415. https://doi.org/10.1002/hbm.25301

Friedman, N. P., & Robbins, T. W. (2022). The role of prefrontal cortex in cognitive control and executive function. *Neuropsychopharmacology, 47*(1), 72–89. https://doi.org/10.1038/s41386-021-01132-0

Goetschius, L. G., Hein, T. C., Mattson, W. I., Lopez-Duran, N., Dotterer, H. L., Welsh, R. C., Mitchell, C., Hyde, L. W., & Monk, C. S. (2019). Amygdala-prefrontal cortex white matter tracts are widespread, variable and implicated in amygdala modulation in adolescents. *NeuroImage, 191*, 278–291. https://doi.org/10.1016/j.neuroimage.2019.02.009

Goetschius, L. G., Hein, T. C., McLanahan, S. S., Brooks-Gunn, J., McLoyd, V. C., Dotterer, H. L., … Beltz, A. M. (2020). Association of Childhood Violence Exposure With Adolescent Neural Network Density. *JAMA Network Open, 3*(9), Article e2017850. https://doi.org/10.1001/jamanetworkopen.2020.17850

Griffiths, K. R., Grieve, S. M., Kohn, M. R., Clarke, S., Williams, L. M., & Korgaonkar, M. S. (2016). Altered gray matter organization in children and adolescents with ADHD: A structural covariance connectome study. *Translational Psychiatry, 6*(11). https://doi.org/10.1038/tp.2016.219. Article 11.

Grönqvist, H., Nilsson, J. P., & Robling, P.-O. (2020). Understanding how low levels of early lead exposure affect children's life trajectories. *Journal of Political Economy, 128*(9), 3376–3433. https://doi.org/10.1086/708725

Hagmann, P., Sporns, O., Madan, N., Cammoun, L., Pienaar, R., Wedeen, V. J., Meuli, R., Thiran, J.-P., & Grant, P. E. (2010). White matter maturation reshapes structural connectivity in the late developing human brain. *Proceedings of the National Academy of Sciences, 107*(44), 19067–19072. https://doi.org/10.1073/pnas.1009073107

Hamilton, O. K. L., Zhang, Q., McRae, A. F., Walker, R. M., Morris, S. W., Redmond, P., Campbell, A., Murray, A. D., Porteous, D. J., Evans, K. L., McIntosh, A. M., Deary, I. J., & Marioni, R. E. (2019). An epigenetic score for BMI based on DNA methylation correlates with poor physical health and major disease in the Lothian birth cohort. *International Journal of Obesity, 43*(9). https://doi.org/10.1038/s41366-018-0262-3. Article 9.

Hardi, F. A., Goetschius, L. G., Peckins, M. K., Brooks-Gunn, J., McLanahan, S. S., McLoyd, V., … Monk, C. S. (2022). Differential Developmental Associations of Material Hardship Exposure and Adolescent Amygdala–Prefrontal Cortex White Matter Connectivity. *Journal of Cognitive Neuroscience, 34*(10), 1866–1891. https://doi.org/10.1162/jocn_a_01801

Hardi, F. A., Goetschius, L. G., Tillem, S., McLoyd, V., Brooks-Gunn, J., Boone, M., Lopez-Duran, N., Mitchell, C., Hyde, L. W., & Monk, C. S. (2023). Early childhood household instability, adolescent structural neural network architecture, and young adulthood depression: A 21-year longitudinal study. *Developmental Cognitive Neuroscience, 61*, Article 101253. https://doi.org/10.1016/j.dcn.2023.101253

Hein, T. C., Mattson, W. I., Dotterer, H. L., Mitchell, C., Lopez-Duran, N., Thomason, M. E., Peltier, S. J., Welsh, R. C., Hyde, L. W., & Monk, C. S (2018). Amygdala habituation and uncinate fasciculus connectivity in adolescence: A multi-modal approach. *NeuroImage, 183*, 617–626. https://doi.org/10.1016/j.neuroimage.2018.08.058

Heiss, J. A., & Just, A. C. (2019). Improved filtering of DNA methylation microarray data by detection p values and its impact on downstream analyses. *Clinical Epigenetics, 11*(1), 15. https://doi.org/10.1186/s13148-019-0615-3

Hsieh, T.-J., Chen, Y.-C., Li, C.-W., Liu, G.-C., Chiu, Y.-W., & Chuang, H.-Y. (2009). A proton magnetic resonance spectroscopy study of the chronic lead effect on the basal ganglion and frontal and occipital lobes in middle-age adults. *Environmental Health Perspectives, 117*(6), 941–945. https://doi.org/10.1289/ehp.0800187

Ingalhalikar, M., Smith, A., Parker, D., Satterthwaite, T. D., Elliott, M. A., Ruparel, K., Hakonarson, H., Gur, R. E., Gur, R. C., & Verma, R. (2014). Sex differences in the structural connectome of the human brain. *Proceedings of the National Academy of Sciences of the United States of America, 111*(2), 823–828. https://doi.org/10.1073/pnas.1316909110

Koo, T. K., & Li, M. Y. (2016). A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *Journal of Chiropractic Medicine, 15*(2), 155–163. https://doi.org/10.1016/j.jcm.2016.02.012

Kragel, P. A., Han, X., Kraynak, T. E., Gianaros, P. J., & Wager, T. D. (2021). Functional MRI can be highly reliable, but it depends on what you measure: A commentary on Elliott et al. (2020). *Psychological Science, 32*(4), 622–626. https://doi.org/10.1177/0956797621989730

Latora, V., & Marchiori, M. (2001). Efficient behavior of small-world networks. *Physical Review Letters, 87*(19), Article 198701. https://doi.org/10.1103/PhysRevLett.87.198701

Lautarescu, A., Craig, M. C., & Glover, V. (2020). Chapter two - prenatal stress: Effects on fetal and child brain development. In A. Clow, & N. Smyth (Eds.), *International review of neurobiology: 150. International review of neurobiology* (pp. 17–40). Academic Press. https://doi.org/10.1016/bs.irn.2019.11.002.

Lebel, C., & Deoni, S. (2018). The development of brain white matter microstructure. *NeuroImage, 182*, 207–218. https://doi.org/10.1016/j.neuroimage.2017.12.097

Li, M., Chen, Z., & Li, T. (2012). Small-world brain networks in schizophrenia. *Shanghai Archives of Psychiatry, 24*(6), 322–327. https://doi.org/10.3969/j.issn.1002-0829.2012.06.003

Liao, X., Vasilakos, A. V., & He, Y. (2017). Small-world human brain networks: perspectives and challenges. *Neuroscience & Biobehavioral Reviews, 77*, 286–300. https://doi.org/10.1016/j.neubiorev.2017.03.018

Llop, S., Lopez-Espinosa, M.-J., Rebagliato, M., & Ballester, F. (2013). Gender differences in the neurotoxicity of metals in children. *Toxicology, 311*(1), 3–12. https://doi.org/10.1016/j.tox.2013.04.015

Ma, T., Chen, H.-H., & Ho, I. K. (1999). Effects of chronic lead (Pb) exposure on neurobehavioral function and dopaminergic neurotransmitter receptors in rats. *Toxicology Letters, 105*(2), 111–121. https://doi.org/10.1016/S0378-4274(98)00388-9

Maloney, B., Bayon, B. L., Zawia, N. H., & Lahiri, D. K. (2018). Latent consequences of early-life lead (Pb) exposure and the future: Addressing the Pb crisis. *NeuroToxicology, 68*, 126–132. https://doi.org/10.1016/j.neuro.2018.06.016

Mansel, C., Fross, S., Rose, J., Dema, E., Mann, A., Hart, H., Klawinski, P., & Vohra, B. P. S (2019). Lead exposure reduces survival, neuronal determination, and differentiation of P19 stem cells. *Neurotoxicology and Teratology, 72*, 58–70. https://doi.org/10.1016/j.ntt.2019.01.005

McFarland, M. J., Hauer, M. E., & Reuben, A. (2022). Half of US population exposed to adverse lead levels in early childhood. *Proceedings of the National Academy of Sciences, 119*(11), Article e2118631119. https://doi.org/10.1073/pnas.2118631119

Menon, V. (2011). Large-scale brain networks and psychopathology: A unifying triple network model. *Trends in Cognitive Sciences, 15*(10), 483–506. https://doi.org/10.1016/j.tics.2011.08.003

Muldoon, S. F., Bridgeford, E. W., & Bassett, D. S. (2016). Small-world propensity and weighted brain networks. *Scientific Reports, 6*(1). https://doi.org/10.1038/srep22057. Article 1.

Muñoz-Moreno, E., Fischi-Gomez, E., Batalle, D., Borradori-Tolsa, C., Eixarch, E., Thiran, J.-P., Gratacós, E., & Hüppi, P. S. (2016). Structural brain network reorganization and social cognition related to adverse perinatal condition from infancy to early adolescence. *Frontiers in Neuroscience, 10*. https://doi.org/10.3389/fnins.2016.00560

Naranjo, V. I., Hendricks, M., & Jones, K. S. (2020). Lead toxicity in children: An unremitting public health problem. *Pediatric Neurology, 113*, 51–55. https://doi.org/10.1016/j.pediatrneurol.2020.08.005

Newman, M. E. J. (2006). Finding community structure in networks using the eigenvectors of matrices. *Physical Review E, 74*(3), Article 036104. https://doi.org/10.1103/PhysRevE.74.036104

Niu, L., Xu, Z., & Taylor, J. A. (2016). RCP: A novel probe design bias correction method for Illumina Methylation BeadChip. *Bioinformatics, 32*(17), 2659–2663. https://doi.org/10.1093/bioinformatics/btw285

Onnela, J.-P., Saramäki, J., Kertész, J., & Kaski, K. (2005). Intensity and coherence of motifs in weighted complex networks. *Physical Review E, 71*(6), Article 065103. https://doi.org/10.1103/PhysRevE.71.065103

R Core Team. (2021). *R: The R project for statistical computing*. Vienna: R Foundation for Statistical Computing. https://www.r-project.org/.

Refn, M. R., Andersen, M. M., Kampmann, M.-L., Tfelt-Hansen, J., Sørensen, E., Larsen, M. H., … Pereira, V. (2023). Longitudinal changes and variation in human DNA methylation analysed with the Illumina MethylationEPIC BeadChip assay and their implications on forensic age prediction. *Scientific Reports, 13*(1), 21658. https://doi.org/10.1038/s41598-023-49064-7

Reichardt, J., & Bornholdt, S. (2006). Statistical mechanics of community detection. *Physical Review E, 74*(1), Article 016110. https://doi.org/10.1103/PhysRevE.74.016110

Reichman, N. E., Teitler, J. O., Garfinkel, I., & McLanahan, S. S. (2001). Fragile families: Sample and design. *Children and Youth Services Review, 23*(4–5), 303–326.

Richmond, R. C., Suderman, M., Langdon, R., Relton, C. L., & Davey Smith, G. (2018). DNA methylation as a marker for prenatal smoke exposure in adults. *International Journal of Epidemiology, 47*(4), 1120–1130. https://doi.org/10.1093/ije/dyy091

Rísová, V. (2019). The pathway of lead through the mother's body to the child. *Interdisciplinary Toxicology, 12*(1), 1–6. https://doi.org/10.2478/intox-2019-0001

Rolls, E. T., Joliot, M., & Tzourio-Mazoyer, N. (2015). Implementation of a new parcellation of the orbitofrontal cortex in the automated anatomical labeling atlas. *NeuroImage, 122*, 1–5. https://doi.org/10.1016/j.neuroimage.2015.07.075

Rubinov, M., & Sporns, O. (2010). Complex network measures of brain connectivity: uses and interpretations. *NeuroImage, 52*(3), 1059–1069. https://doi.org/10.1016/j.neuroimage.2009.10.003

Rygiel, C. A., Goodrich, J. M., Solano-González, M., Mercado-García, A., Hu, H., Téllez-Rojo, M. M., Peterson, K. E., & Dolinoy, D. C. (2021). Prenatal lead (Pb) exposure and peripheral blood DNA methylation (5mC) and hydroxymethylation (5hmC) in Mexican adolescents from the ELEMENT birth cohort. *Environmental Health Perspectives, 129*(6), Article 067002. https://doi.org/10.1289/EHP8507

Senut, M.-C., Cingolani, P., Sen, A., Kruger, A., Shaik, A., Hirsch, H., Suhr, S., & Ruden, D. (2012). Epigenetic of early-life lead exposure and effects on brain development. *Epigenomics, 4*, 665–674. https://doi.org/10.2217/epi.12.58

Senut, M.-C., Sen, A., Cingolani, P., Shaik, A., Land, S. J., & Ruden, D. M. (2014). Lead exposure disrupts global DNA methylation in Human embryonic stem cells and alters their neuronal differentiation. *Toxicological Sciences, 139*(1), 142–161. https://doi.org/10.1093/toxsci/kfu028

Sha, Z., Schijven, D., Fisher, S. E., & Francks, C. (2023). Genetic architecture of the white matter connectome of the human brain. *Science Advances, 9*(7), eadd2870. https://doi.org/10.1126/sciadv.add2870

Siugzdaite, R., Bathelt, J., Holmes, J., & Astle, D. E. (2020). Transdiagnostic brain mapping in developmental disorders. *Current Biology, 30*(7), 1245–1257. https://doi.org/10.1016/j.cub.2020.01.078. e4.

Song, L., Mishra, V., Ouyang, M., Peng, Q., Slinger, M., Liu, S., & Huang, H. (2017). Human fetal brain connectome: structural network development from middle fetal stage to birth. *Frontiers in Neuroscience, 11*. https://doi.org/10.3389/fnins.2017.00561

Taylor, P. N., Wang, Y., & Kaiser, M. (2017). Within brain area tractography suggests local modularity using high resolution connectomics. *Scientific Reports, 7*(1), Article 39859. https://doi.org/10.1038/srep39859

Taxali, A., Angstadt, M., Rutherford, S., & Sripada, C. (2021). Boost in test-retest reliability in resting State fMRI with predictive modeling. *Cerebral Cortex, 31*(6), 2822–2833. https://doi.org/10.1093/cercor/bhaa390 (New York, N.Y.: 1991).

Teye, S. O., Yanosky, J. D., Cuffee, Y., Weng, X., Luquis, R., Farace, E., & Wang, L. (2021). Exploring persistent racial/ethnic disparities in lead exposure among American children aged 1–5 years: Results from NHANES 1999–2016. *International Archives of Occupational and Environmental Health, 94*(4), 723–730. https://doi.org/10.1007/s00420-020-01616-4

Thomason, M. E. (2020). Development of brain networks In utero: relevance for common neural disorders. *Biological Psychiatry, 88*(1), 40–50. https://doi.org/10.1016/j.biopsych.2020.02.007

Thomason, M. E., Hect, J. L., Rauh, V. A., Trentacosta, C., Wheelock, M. D., Eggebrecht, A. T., Espinoza-Heredia, C., & Burt, S. A. (2019). Prenatal lead exposure impacts cross-hemispheric and long-range connectivity in the human fetal brain. *NeuroImage, 191*, 186–192. https://doi.org/10.1016/j.neuroimage.2019.02.017

Tournier, J.-D., Calamante, F., & Connelly, A. (2007). Robust determination of the fibre orientation distribution in diffusion MRI: Non-negativity constrained super-resolved

spherical deconvolution. *NeuroImage, 35*(4), 1459–1472. https://doi.org/10.1016/j.neuroimage.2007.02.016

Tournier, J.-D., Calamante, F., Gadian, D. G., & Connelly, A. (2004). Direct estimation of the fiber orientation density function from diffusion-weighted MRI data using spherical deconvolution. *NeuroImage, 23*(3), 1176–1185. https://doi.org/10.1016/j.neuroimage.2004.07.037

Tung, P. W., Kennedy, E. M., Burt, A., Hermetz, K., Karagas, M., & Marsit, C. J. (2022). Prenatal lead (Pb) exposure is associated with differential placental DNA methylation and hydroxymethylation in a human population. *Epigenetics, 17*(13), 2404–2420. https://doi.org/10.1080/15592294.2022.2126087

Varma, G., Sobolewski, M., Cory-Slechta, D. A., & Schneider, J. S. (2017). Sex- and brain region- specific effects of prenatal stress and lead exposure on permissive and repressive post-translational histone modifications from embryonic development through adulthood. *NeuroToxicology, 62*, 207–217. https://doi.org/10.1016/j.neuro.2017.07.002

Wang, Y., Ghumare, E., Vandenberghe, R., & Dupont, P. (2017). Comparison of different generalizations of clustering coefficient and local efficiency for weighted undirected graphs. *Neural Computation, 29*(2), 313–331. https://doi.org/10.1162/NECO_a_00914. Neural Computation.

Wani, A. L., Ara, A., & Usmani, J. A. (2015). Lead toxicity: A review. *Interdisciplinary Toxicology, 8*(2), 55–64. https://doi.org/10.1515/intox-2015-0009

Watts, D. J., & Strogatz, S. H. (1998). Collective dynamics of "small-world" networks. *Nature, 393*(6684), 440–442. https://doi.org/10.1038/30918

Wheeler, W., & Brown, M. J. (2013). Blood lead levels in children aged 1–5 years—United States, 1999–2010. *Morbidity and Mortality Weekly Report, 62*(13), 245–248.

White, L. D., Cory-Slechta, D. A., Gilbert, M. E., Tiffany-Castiglioni, E., Zawia, N. H., Virgolini, M., Rossi-George, A., Lasley, S. M., Qian, Y. C., & Basha, Md. R. (2007). New and evolving concepts in the neurotoxicology of lead. *Toxicology and Applied Pharmacology, 225*(1), 1–27. https://doi.org/10.1016/j.taap.2007.08.001

Winter, A. S., & Sampson, R. J. (2017). From lead exposure in early childhood to adolescent health: A Chicago birth cohort. *American Journal of Public Health, 107*(9), 1496–1501. https://doi.org/10.2105/AJPH.2017.303903

Wu, B., Chen, Y., Long, X., Cao, Y., Xie, H., Wang, X., Roberts, N., Gong, Q., & Jia, Z. (2023). Altered single-subject gray matter structural networks in first-episode drug-naïve adolescent major depressive disorder. *Psychiatry Research, 329*, Article 115557. https://doi.org/10.1016/j.psychres.2023.115557

Wu, S., Hivert, M.-F., Cardenas, A., Zhong, J., Rifas-Shiman, S. L., Agha, G., Colicino, E., Just, A. C., Amarasiriwardena, C., Lin, X., Litonjua, A. A., DeMeo, D. L., Gillman, M. W., Wright, R. O., Oken, E., & Baccarelli, A. A. (2017). Exposure to low levels of lead in utero and umbilical cord blood DNA methylation in project Viva: An epigenome-wide association study. *Environmental Health Perspectives, 125*(8), Article 087019. https://doi.org/10.1289/EHP1246

Xu, Z., Niu, L., Li, L., & Taylor, J. A. (2016). ENmix: A novel background correction method for Illumina HumanMethylation450 BeadChip. *Nucleic Acids Research, 44*(3), e20. https://doi.org/10.1093/nar/gkv907

Yang, F., Zhang, J., Fan, L., Liao, M., Wang, Y., Chen, C., Zhai, T., Zhang, Y., Li, L., Su, L., & Dai, Z. (2020). White matter structural network disturbances in first-episode, drug-naïve adolescents with generalized anxiety disorder. *Journal of Psychiatric Research, 130*, 394–404. https://doi.org/10.1016/j.jpsychires.2020.08.004